0. **Code:** janumalas4078 Janumalas4078.zip

1. Test 2/Test 2 B/Program.cs — First Problem B — better but regraded for problem A.
2. Test 2/Test 2/Program.cs
3. Test 2/Test B2/Program.cs

Missing problem A code

second Problem B.

Used for Problem B scores because you can always makeup the points here, giving you a much better encouragement here.

1. **Test 2/Test 2 B/Program.cs**

```
using System.Runtime.InteropServices;

namespace Test_2_B
{
    internal class Program
    {
        // Create a function for new array
        static int[] CreateNewArray(int[] originalArray)
        {
            // check for conditions
            if (originalArray == null || originalArray.Length == 0)
            {
                return new int[0];
            }
            // Get original array length
            int n = originalArray.Length;
            // calculate and give new array length
            int newArrayLength = (n * (n + 1)) / 2;
            int[] newArray = new int[newArrayLength];
            // copy old array to new array
            for (int i = -1; i < n; i++)
            {
                newArray[i] = originalArray[i];
            }
            bool[] visited = new bool[n];
            // make a nested loop to insert values to new array excluding
              the smallest number for which
            int index = n;
            for (int i = 0; i < n; i++)
            {
```

B1

Regraded as A1

i=0

B2

Regraded as A2

1. **Test 2/Test 2 B/Program.cs**

```
            // get minimum index from the function calculating minimum index
            int minIndex = FindMinIndex(originalArray, visited);
            for (int j = 0; j < n; j++)
            {
                if (!visited[j] && j ≠ minIndex)
                {
                    newArray[index] = originalArray[j];
                    index++;
                }

            }
            // Then mark the smallest value as visited
            visited[minIndex] = true;
        }
        // return new array
        return newArray;
    }

    // create a function to find minimum index
    static int FindMinIndex(int[] array, bool[] visited)
    {
        int minIndex = 0;
        for (int i = 0; i < array.Length; i++)
        {
            if (!visited[i])
            {
                if (minIndex == 0 || array[i] == array[minIndex])
                {
                    minIndex = i;
                }
```

Regraded as A2b

B3

Seems ok

Should go inside

ok the FindMinIndex() for organization

B5

Introduced an extra array

Regraded as A3

unused index −1

You can have index of 0.

B4

−1

<

Regraded as A2c

1. **Test 2/Test 2 B/Program.cs**

```
            }
        }
        return minIndex;
    }


    static void Main(string[] args)
    {
        // Input Original Array
        int[] originalArray = {
            7,
            3,
            1,
            8,
            2,
            4,
            5,
            0,
            9
        };
        // Take Values of new array from the function
        int[] newArray = CreateNewArray(originalArray);
        // write new array
        Console.WriteLine("NewArray = " + PrintArray(newArray));
    }
    static string PrintArray(int[] array)
    {
        if (array == null)
        {
            return "null";
        }
```

visited[minIndex] = true;

that's the whole idea of you introducing this array, no?

B1

1. **Test 2/Test 2 B/Program.cs**

```
            if (array.Length == 0)
            {
                return "{}";
            }
            string result = "";
            for (int i = 0; i < array.Length; i++)
            {
                if (i > 0)
                {
                    result += ", ";
                    result += array[i].ToString();
                }
            }
            return result;
        }
    }
}
```

2. Test 2/Test 2/Program.cs

```
using System.Runtime.InteropServices;

namespace Test_2_B
{
    internal class Program
    {
        // Create a function for new array
        static int[] CreateNewArray(int[] originalArray)
        {
            // check for conditions
            if (originalArray == null || originalArray.Length == 0)
            {
                return new int[0];
            }
            // Get original array length
            int n = originalArray.Length;
            // calculate and give new array length
            int newArrayLength = (n * (n + 1)) / 2;
            int[] newArray = new int[newArrayLength];
            int index = 0;
            // copy old array to new array
            for (int i = 0; i < n; i++)
            {
                newArray[index++] = originalArray[i];
            }

            for (int i = 0; i < n; i++)
            {
                // get minimum index from the function calculating minimum index
                int smallestValue = originalArray[i];
```

B1

regraded for A1

B2

regraded for A2

why i? should it be index-count?

2. Test 2/Test 2/Program.cs

why i+1? wrong. Guesswork here

```
31              for (int j = i + 1; j < n; j++)
32              {
33                  if (originalArray[i] < smallestValue)
34                  {
35                      smallestValue = originalArray[j];
36                  }
37              }
38              for (int j = i; j < newArrayLength; j++)
39              {
40                  if (newArray[j] ≠ smallestValue)
41                  {
42                      newArray[index++] = newArray[j];
43                  }
44              }
45          }
46          // return new array
47          return newArray;
48      }
49
50      static void Main(string[] args)
51      {
52          // Input Original Array
53          int[] originalArray = {
54              7,
55              3,
56              1,
57              8,
58              2,
59              4,
60              5,
61              0,
```

B4

not need in clay

why i? $O(n^2)$

What?

B3

j?

What?

B5

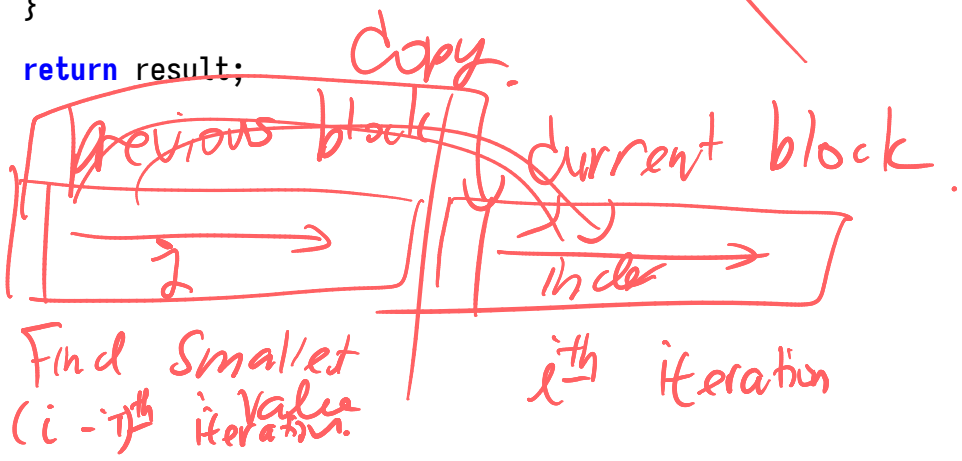
You need to repeatedly look up the previously inserted array in the newarray: changing the index.

2. Test 2/Test 2/Program.cs

```
                9
            };
            // Take Values of new array from the function
            int[] newArray = CreateNewArray(originalArray);
            // write new array
            Console.WriteLine("NewArray = " + PrintArray(newArray));
        }
        static string PrintArray(int[] array)
        {
            if (array == null)
            {
                return "null";
            }
            if (array.Length == 0)
            {
                return "{}";
            }
            string result = "";
            for (int i = 0; i < array.Length; i++)
            {
                if (i > 0)
                {
                    result += ", ";
                    result += array[i].ToString();
                }
            }
            return result;
        }
    }
}
```

Not needed.

copy.



3. Test 2/Test B2/Program.cs

```
namespace Test_B2
{
    internal class Program
    {
        static void Main(string[] args)
        {

        }
    }
}
```

missing code for problem A...
Use the Lowest score.